

## SAFETY DATA SHEET

Creation Date 03-December-2010

Revision Date 28-March-2024

Revision Number 3

### 1. Identification

**Product Name** Magnesium oxide substrate

**Cat No. :** 38503

**CAS-No** 1309-48-4  
**Synonyms** Magnesia.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component       | CAS-No    | Weight % |
|-----------------|-----------|----------|
| Magnesium oxide | 1309-48-4 | 98       |

#### 4. First-aid measures

|                                                               |                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                                             | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention immediately if symptoms occur.       |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Get medical attention.                                                                          |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | No information available.<br>Treat symptomatically                                                                      |

#### 5. Fire-fighting measures

|                                         |                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                     |
| <b>Flash Point</b>                      | No information available                                                     |
| <b>Method -</b>                         | No information available                                                     |
| <b>Autoignition Temperature</b>         | No information available                                                     |
| <b>Explosion Limits</b>                 |                                                                              |
| <b>Upper</b>                            | No data available                                                            |
| <b>Lower</b>                            | No data available                                                            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                     |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                     |

##### Specific Hazards Arising from the Chemical

Non-combustible. Containers may explode when heated.

##### Hazardous Combustion Products

Magnesium oxides.

##### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

##### NFPA

**Health**  
1

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                  |                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. |
| <b>Environmental Precautions</b> | See Section 12 for additional Ecological Information.                                             |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

#### 7. Handling and storage

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin and eyes. Do not breathe dust. |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Water.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component       | Alberta                   | British Columbia                                                                    | Ontario TWAEV             | Quebec                    | ACGIH TLV                 | OSHA PEL                                                         | NIOSH                       |
|-----------------|---------------------------|-------------------------------------------------------------------------------------|---------------------------|---------------------------|---------------------------|------------------------------------------------------------------|-----------------------------|
| Magnesium oxide | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> | (Vacated) TWA: 10 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup> | IDLH: 750 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations | -               | Splash protection only |
| Nitrile rubber |                                   |                 |                        |
| Neoprene       |                                   |                 |                        |
| PVC            |                                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

**Physical State**

Powder Solid

**Appearance**

White

**Odor**

Odorless

**Odor Threshold**

No information available

**pH**

10.3 saturated solution

|                                        |                          |
|----------------------------------------|--------------------------|
| Melting Point/Range                    | 2800 °C / 5072 °F        |
| Boiling Point/Range                    | 3600 °C / 6512 °F        |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 0 hPa @ 20 °C            |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 3.580                    |
| Solubility                             | Hydrolyses               |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Mg O                     |
| Molecular Weight                       | 40.29                    |

## 10. Stability and reactivity

|                                  |                                                                         |
|----------------------------------|-------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                              |
| Stability                        | Unstable on exposure to moisture. Moisture sensitive.                   |
| Conditions to Avoid              | Incompatible products. Exposure to air. Exposure to moist air or water. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Water                            |
| Hazardous Decomposition Products | Magnesium oxides                                                        |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                |
| Hazardous Reactions              | None under normal processing.                                           |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

#### Component Information

| Component       | LD50 Oral                                              | LD50 Dermal | LC50 Inhalation |
|-----------------|--------------------------------------------------------|-------------|-----------------|
| Magnesium oxide | LD50 = 3990 mg/kg ( Rat )<br>LD50 = 3870 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component       | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------|-----------|------------|------------|------------|------------|------------|
| Magnesium oxide | 1309-48-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

|                                                   |                                                                |
|---------------------------------------------------|----------------------------------------------------------------|
| <b>Developmental Effects</b>                      | No information available.                                      |
| <b>Teratogenicity</b>                             | No information available.                                      |
| <b>STOT - single exposure</b>                     | None known                                                     |
| <b>STOT - repeated exposure</b>                   | None known                                                     |
| <b>Aspiration hazard</b>                          | No information available                                       |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                       |
| <b>Endocrine Disruptor Information</b>            | No information available                                       |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. |

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

|                                      |                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Soluble in water Persistence is unlikely based on information available.                                       |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                                                                      |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. Is not likely mobile in the environment. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

### International Inventories

| Component       | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Magnesium oxide | 1309-48-4 | X   | -    | X    | ACTIVE                                        | 215-171-9 | -      | -   |

| Component       | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Magnesium oxide | 1309-48-4 | X     | KE-22728 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component       | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Magnesium oxide | 1309-48-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component       | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Magnesium oxide | 1309-48-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

### 16. Other information

**Prepared By**

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

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**End of SDS**